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Optimizing feed frame design and tableting process parameters to increase die-filling uniformity on a high-speed rotary tablet press

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ABSTRACT

Despite the high quantities of tablets produced daily, many tableting processes are still operated at sub-optimal settings and hence lack the necessary flexibility to mitigate for possible process deviations. However, to ensure this flexibility on tableting throughput it is important to select the most robust feed frame design and settings regarding die-filling. In this research study, four paddle designs for a two-compartment forced feeder (equipped with a metering and a feeding paddle wheel) were evaluated at a wide range of process-settings (i.e. tableting speed, paddle speed, overfill level) and feed frame features (i.e. deaeration) for their impact on the die-filling step of a poorly flowing model formulation (i.e. MCC 101) using a quality-by-design approach. No benefit on die-filling was observed when using higher speeds of the metering paddle wheel compared to the feeding paddle wheel, and no convincing arguments were obtained to use the feed frame deaeration opening. Some combinations of paddle design and process-settings significantly increased the risk for inconsistent die-filling (i.e. high tablet weight variability) which can therefore limit the efficiency of the tableting process. The approach used in this study enabled to compare the paddle designs for their die-filling performance in function of varying tableting speeds, eventually resulting in the selection of a feed frame design that is most robust and therefore will provide a uniform die-filling over a wide range of throughputs. Selection of the most robust parameters is an important prerequisite for the ability of using the rotary tablet press as an agile unit-operation.

1. Introduction

Powder compression on rotary tablet presses is widespread in various industries, which is clearly reflected in pharmaceuticals where tablets are by far the most popular dosage form. The ease of manufacturing combined with a high economic efficiency, high accuracy of dosing and patient convenience all contribute to the dominance of tablets (Armstrong, 2007). Die-filling can be considered as a crucial control variable of the tableting process since it is critical for obtaining the desired tablet weight and hence drug dose, but also less obvious quality attributes such as size, tensile strength and drug release are (in) directly affected by this step (Armstrong, 2007; Mateo-Ortiz and Méndez, 2015; Sinka et al., 2009; Xie and Puri, 2006). Moreover, die-filling can be rate-limiting for the tableting process and therefore impact the throughput of the entire continuous line when the tablets press in linked to other unit-operations.

With the pharmaceutical industry gradually shifting towards continuous manufacturing, the use of a rotary tablet press in continuous manufacturing lines is often considered because of its inherently continuous nature (Byrn et al., 2015). While a tablet press in batch processing is mainly operated at a constant throughput, this operation mode could create problems in a continuous production line (e.g. material holdup at the tablet press or needing to shut down the press) if mass flow variation occurs in the upstream unit operations of the line (Martinetz et al., 2017). Ideally, each unit operation should be able to adapt to changes in throughput without negatively impacting the

Abbreviations: API, active pharmaceutical ingredient; CP, centerpoint; CQA, critical quality attribute; DOE, design of experiments; MBPH, main-compression bottom punch height; MCC, microcrystalline cellulose; MCD, main-compression displacement; MCF, main compression force; MLR, multiple linear regression; Pa1, feeding paddle wheel (1); Pa2, metering paddle wheel (2); PBPH, pre-compression bottom punch height; PC, principal component; PCA, principal component analysis; PCD, pre-compression displacement; PCF, pre-compression force; QbD, quality-by-design; RSD, residual standard deviation; s, seconds; TH, tablet hardness; TW, tablet weight

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quality of intermediates or final products. Martinetz et al. recently introduced an interesting concept to step away from the conventional view of the rotary tablet press as the so called “master” in a control concept since it defined the mass flow through the entire process by operating at constant turret speed. They proposed an operation concept which introduces flexibility in the process to handle upstream mass flow fluctuations by modulating the press turret speed in function of the hopper fill level and hence adjusting the mass flow rate into and from the tablet press (Martinetz et al., 2017). This concept was verified for good flowing mixtures and proved that the tablets critical quality attributes (CQAs) were not significantly affected within the proposed design space.

However, it is generally recognized that higher turret speeds have a negative impact on the die filling step, especially when poorly flowing powders are used (Mendez et al., 2010; Yaginuma et al., 2007). Pharmaceutical powders often lack sufficient flow properties and flowability of powders is often the rate limiting step during die-filling. In most high-speed rotary tablet presses, feed frames are used to facilitate filling of the die-cavities with powder. During this die-filling step, various mechanisms (i.e. gravity feeding, forced feeding, suction fill, overhead pressure, centrifugal forces, air pressure and vibrations) act simultaneously (Jackson et al., 2007; Peeters et al., 2015; Schneider et al., 2007; Sinka et al., 2004). Particularly poorly flowing powders are more subjected to forced feeding than gravity feeding and hence it is expected that the feed frame design will play a major role in the die-filling step of these challenging formulations (Mendez et al., 2010; Peeters et al., 2015; Sun, 2010). A deliberately selected, robust paddle design for the feed frame could guarantee a uniform die-filling step for such formulations at fluctuating turret speeds when the rotary tablet press is used as agile unit-operation in a continuous manufacturing line.

Although a wide variety of feed frames which differ in number of compartments, paddle-design and volume are currently available, up till now there is only limited public knowledge on when and how to use a specific design at its most suitable process settings to obtain the best tablet quality. This explains the rather paradoxical situation that many companies and institutions currently face: despite the fact that tablets are already produced for decades, the established tableting process are still too often based on trial-and-error or ‘in-house’ knowledge and many tablet presses are operated at sub-optimal settings.

The aim of this research paper is therefore to provide in-depth knowledge on the various paddle-designs of a two-compartment forced feeder and their impact on the die-filling step of a rotary tablet press under realistic process-settings. This knowledge could contribute to an improved set-up of industrial tableting processes by choosing the most robust feed frame design for a wide range of tableting process-settings.

2. Materials and methods

Microcrystalline cellulose (Vivapur® 101, JRS Pharma, Rosenberg, Germany) was selected as model powder based on a raw material database (Van Snick et al., 2018) which included a wide variety of characteristics. No lubrication or pre-treatment of MCC was performed prior to the tabletting experiments.

2.1. Experimental tableting set-up

All experiments were performed on a rotary tablet press (MODUL™ P, GEA Process Engineering, Halle, Belgium). The powder flows from the material hopper by means of a rotary valve (controlled via a level-sensor) in the feed tube to finally enter the feed frame (Fig. 1). The level sensor ensures a constant fill level of the feed tube in order to guarantee a constant overhead pressure on the powder in the feed frame throughout all tableting runs. The turret was mounted with 10 cylindrical flat-faced Euro B punch pairs, while the fill-depth was fixed at 8 mm for all experiments. To perform experiments at different overfill levels of the experimental design, the overfill cams were switched

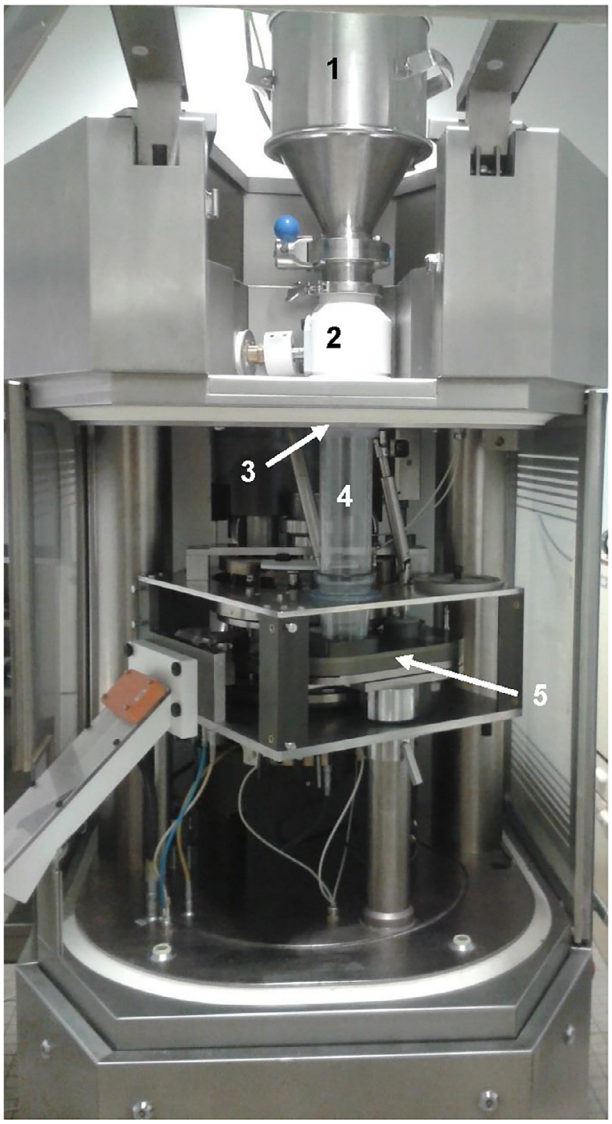


Fig. 1. Experimental set-up of the GEA Modul™ P tablet press: (1) material hopper, (2) rotary valve, (3) level sensor on (4) plexiglass feed tube and (5) feed frame.

between 8, 10 or 12 mm. For each experiment, the punch positions were adjusted to reach an average pre-compression displacement (PCD), pre-compression force (PCF) and main compression force (MCF) of 0.5 mm, 2 kN and 15 kN, respectively, while no main-compression displacement (MCD) was allowed. The MultiControl 5 software package (GEA Process Engineering, Belgium) controlled the Modul™ P and logged a wide variety of in-line process responses (e.g. PCD, MCF, punch positions) at 1 s intervals. The tablet sampling station was connected to an external balance (K-sampler, Coperion K-tron, Niederlenz, Switzerland), logging the weight gain every second to determine the steady state phase based on the tablet output. All available control loops of the press (i.e. weight, force or displacement control) were deactivated to assure a correct assessment of the die-filling step, which would otherwise be influenced by the compensating control systems.

2.2. Feed frame design

A conventional two-compartment feed frame was used which consists of a top plate with a sealable deaeration opening, two coplanar paddle wheels and a base plate with a slit where the dies pass (Fig. 2). Powder is delivered into the feed frame by an opening above the first,

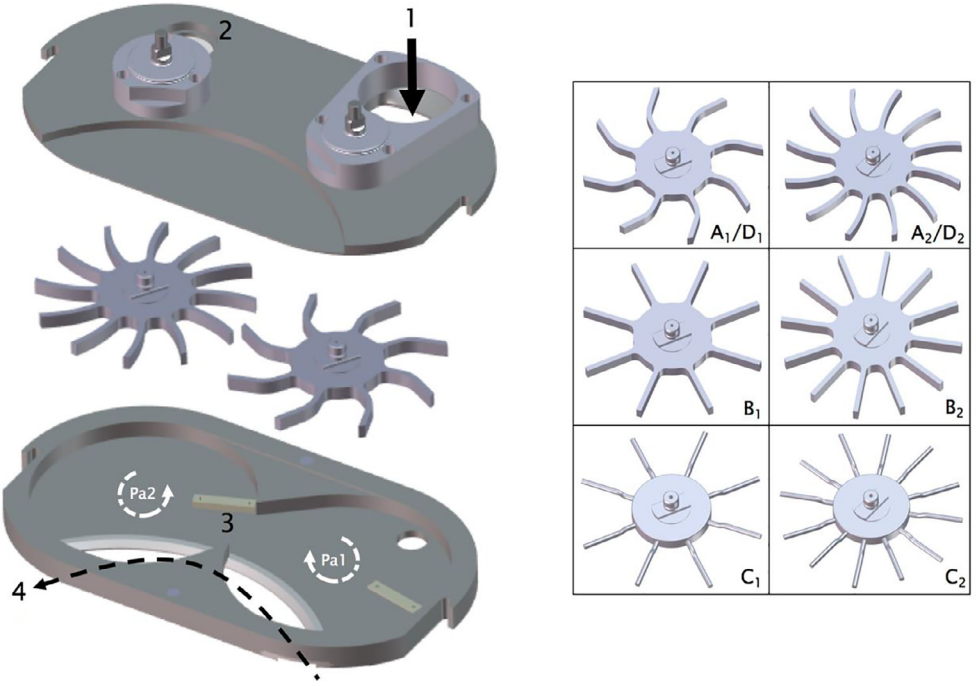


Fig. 2. Overview of the used feed frame (left-hand side) existing of: a top plate with a feed (1) and deaeration (2) opening, two paddles (Pa1, Pa2) with indication of their rotation and a bottom plate with slit for the passing dies (4). At the right side, the different paddle designs (A–D) used for this study are shown with the numbers referring to the use as Pa1 or Pa2. In case of paddle type D the paddle height is reduced compared to paddle A while scrapers are mounted on specific locations (3) at the top plate (Courtesy of GEA Process Engineering).

'feeding' paddle wheel (Pa1), which consists of eight paddles to force the powder towards the dies in a clockwise direction. The second, 'metering' wheel (Pa2), operating in a counter-clockwise direction, has twelve paddles to recover the excess of powder ejected by the weight adjustment unit and returning it to the feeding wheel. Four paddle designs were evaluated in this project which differed by paddle shape and paddle positioning in the wheel (Fig. 2). They were selected based on their market prevalence and assembled by GEA Process Engineering (Halle, Belgium).

In a first part of the study, four paddle pairs (in this paper referred to as paddle A, B, C and D) were evaluated for an eight-fingered Pa1 and a twelve-fingered Pa2. The first paddle pair (A) is a widespread design that is by default used in the forced feeder of this tablet press, consisting of radial curved cuboid paddles for which the curvature is believed to facilitate powder transportation. A second paddle design that was investigated in this study, a paddle wheel with straight cuboid paddles (B), can also be considered as a standard paddle wheel for a wide variety of tableting applications. Paddle type C was designed to improve the 'aeration' of cohesive powders and is characterized by fine cylindrical paddles which are implemented at varying heights in the paddle wheel. It is believed that by this specific implementation of these fine paddles, powder beds are more agitated and hence aerated to improve the flow of cohesive powders which could positively affect the die-filling. The last evaluated paddle design (D) has a similar radial curvature in their cuboid paddles as type A but is combined with a 'scraper' for each paddle wheel, mounted at well-defined positions in the top plate of the feed frame (Fig. 2). The rationale for designing these scrapers was to increase powder bed aeration without the need for changing from cuboid to cylindrical paddles (i.e. reduction in total paddle surface available for power transportation). When scrapers are mounted on the top plate, a partial volume is available on top of the powder bed after paddle passage which could enable more aeration of the powder bed. However, because of the presence of this scrapers the paddles had to be lowered in height compared to type A.

2.3. Design of experiments

Besides the impact of feed frame design, it was intended to understand how process settings might influence the die-filling step. A D-optimal design was selected to investigate the impact of qualitative factors: paddle design (A, B, C & D), deaeration (with/without) together with varying quantitative factors; turret speed (10–100 rpm), overfill level (0–4 mm), speed of the feeding paddle Pa1 (15–100 rpm) and the speed ratio of the metering paddle Pa2 over the feeding paddle Pa1 (1–1.4) on a wide variety of product and process responses. The paddle speed ratio above 1 guaranteed that the speed of Pa1 never exceeded the speed of Pa2, which is important to avoid overfilling and powder compaction in the forced feeder (Peeters et al., 2015). All factor ranges from the design of experiments (DOE) were selected based on preliminary experiments and by taking into account the operational ranges of the tablet press. An experimental design (Table 1) with 48 experiments, including eight centerpoints (two for each paddle design) was constructed. The results were analysed using MODDE 12.0 software (Umetrics, Umeå, Sweden). Multiple linear regression (MLR) was used for calculation of the regression models for each response, where all factors were scaled and centered making the regression coefficients comparable for the different factors. 95% confidence intervals were calculated for all regression coefficients in order to evaluate if factors or factor interactions were significant (i.e. 95% confidence interval of the corresponding regression coefficient not including zero).

2.4. Experimental procedure

Before sampling, the press was run until the system's steady state (i.e. stable mass balance of the unit operation) was reached, generally in less than two minutes for all experiments. Afterwards, twenty grab samples were taken based on a predefined time-scale. The duration and interval of sampling for each run was calculated to guarantee a similar number of tablets for sample analysis, based on the corresponding turret speed and the operator feasibility: every 80 s, a sampling of 20 s was performed for the runs at a tableting speed of 10 rpm, while 2 s sampling every 8 s was sufficient for runs at 100 rpm. At centerpoint (CP) settings, a sample was taken during 4 s every 20 s. This sampling strategy resulted in total run times of 32, 7.67 and 3.2 min after reaching steady state at 10, 55 and 100 rpm turret speeds, respectively.

Table 1
Overview of the performed tableting experiments performed at the settings of the investigated design space with the experiments in bold representing the excluded runs.

Run	Paddle Type	Overfill (mm)	Turret speed (rpm)	Speed Pa1 (rpm)	Speed Pa2 (rpm)	Deaeration
1	A	4	100	15	15	Yes
2	A	4	10	100	140	Yes
3	A	4	10	100	100	No
4	A	4	10	15	21	No
5	A	4	100	100	140	No
6	B	4	10	15	15	Yes
7	B	4	100	100	100	Yes
8	B	4	100	15	21	Yes
9	B	4	100	15	15	No
10	B	4	10	100	140	No
11	C	4	100	15	15	Yes
12[a]	C	4	10	100	100	Yes
13	C	4	10	15	21	Yes
14	C	4	10	15	15	No
15	C	4	100	100	140	No
16[b]	D	4	10	15	15	Yes
17	D	4	100	100	140	Yes
18	D	4	100	100	100	No
19[b]	D	4	100	15	21	No
20	D	4	10	100	140	No
21	A	0	10	15	15	Yes
22	A	0	100	100	100	Yes
23	A	0	100	15	21	Yes
24	A	0	100	15	15	No
25	A	0	10	100	140	No
26	B	0	100	15	15	Yes
27	B	0	10	100	140	Yes
28	B	0	10	100	100	No
29	B	0	10	15	21	No
30	B	0	100	100	140	No
31	C	0	10	15	15	Yes
32	C	0	100	100	140	Yes
33	C	0	100	100	100	No
34	C	0	100	15	21	No
35[a]	C	0	10	100	140	No
36[b]	D	0	100	15	15	Yes
37	D	0	10	100	100	Yes
38[b]	D	0	10	15	21	Yes
39[b]	D	0	10	15	15	No
40	D	0	100	100	140	No
41	A	2	55	57.5	69	No
42	A	2	55	57.5	69	No
43	B	2	55	57.5	69	No
44	B	2	55	57.5	69	No
45	C	2	55	57.5	69	No
46	C	2	55	57.5	69	No
47	D	2	55	57.5	69	No
48	D	2	55	57.5	69	No

[a] Run 12 and 35 were excluded from the experimental design because of extreme die-filling variability.

[b] Run 16, 19, 36, 38 and 39 were excluded from the experimental design because of feeder blockage.

2.5. Product & process responses

2.5.1. Tablet weight

The tablet weight (TW) is directly linked with the die-filling performance in this study since the weight adjustment control loop was switched off and the overfill level remained fixed throughout each run (i.e. open loop). Therefore, the mean TW could be used as a response for the die-filling performance. A combitester (Multitest 50, Dr. Schleuniger, Sotax, Basel, Switzerland) was used to measure TW.

2.5.2. Tablet weight variability

The overall TW variability can be due to drifts or oscillations in mean TW during processing or to inherent variability in die filling at any random point in the process. Calculation of the relative standard deviation of the mean TW of twenty grab-samples (RSD TW_{mean}; Eq. (1)) provided information on the first cause of TW variability (i.e. drifts or oscillation).

$$RSD\ TW_{Mean}\ (\%) = 100 \times \frac{\sigma_{TW\ S1;\ldots;TW\ S20}}{\mu_{TW\ S1;\ldots;TW\ S20}} \quad (1)$$

In Eq. (1), $\sigma_{TW\ S1;\ldots;TW\ S20}$ represents the standard deviation of the mean tablet weights for the twenty grab samples while $\mu_{TW\ S1;\ldots;TW\ S20}$ represents the mean TW of the same grab samples. Moreover, each grab sample can be considered as a random subgroup and the variability herein can be estimated by calculation of the pooled residual standard deviation (RSD TW_{sample}; Eq. (2)).

$$RSD\ TW_{Sample}\ (\%) = 100 \times \frac{\sigma_{pooled}}{\mu_{pooled}} \quad (2)$$

2.5.3. Tablet hardness

Variability on hardness or breaking force of the tablets (RSD TH) was an indirect result of the die-filling step in this study, since no displacement of the main compression roller was allowed. TH was measured using a combitester (Multitest 50, Dr. Schleuniger, Sotax, Basel, Switzerland) and the variability was calculated based on Eq. (1).

2.5.4. Bottom punch heights

Before each experiment, the heights for the bottom punch at both the pre-compression station (PBPH) and main-compression station (MBPH) were modified to reach a fixed PCD, PCF and MCF since a constant fill-depth was maintained. Therefore, PBPH and MBPH was dependent of the amount of material fed inside the dies and these values were kept fixed throughout each run by deactivating the control-loops.

2.5.5. Variability on force and displacement

Performing the tableting experiments in 'open loop' enabled to monitor the variability on PCD and MCF, since they directly depend on the filling of the dies when fixed PBPH and MBPH are maintained during each run. These in-line measured process responses were logged by the MultiControl 5 software, while the RSD was calculated based on Eq. (1).

2.6. Multivariate data analysis

Principal component analysis (PCA) was executed on the relevant data in order to classify the DOE-experimental runs according to their impact on product and process responses using the multivariate data analysing software SIMCA 15.0 (Umetrics, Umeå, Sweden). The dataset consisted of 48 rows (i.e. the DOE-experiments) for the 9 variables (i.e. DOE responses). PCA is a multivariate projection method which extracts and displays the variation in the data set [30]. The data was pre-processed by unit variate scaling and centered in order to balance the weight of each variable.

3. Results and discussion

3.1. Selection of model powder

In this study, the selection of a well-considered model powder was essential since the importance of powder flow properties during die-filling has been illustrated by several research groups (Mendez et al., 2010; Peeters et al., 2015; Sun, 2010). When selecting a free-flowing powder, a less pronounced impact of feed frame design and its process settings is expected on the die-filling step since free-flowing powders are less subjected to forced feeding compared to poorly flowing powders. Moreover, many active pharmaceutical ingredients (API) face flowability problems, in some cases even after formulation

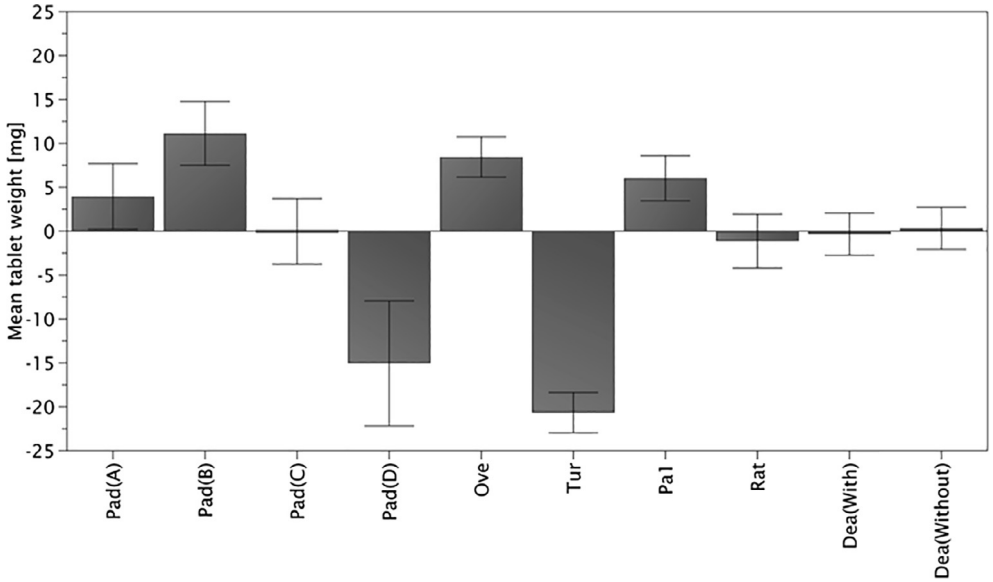


Fig. 3. Coefficient plot for mean tablet weight; regression coefficients including 95% confidence interval for the DOE-factors: paddle type (Pad A–D), overfilling (Ove), turret speed (Tur), paddle speed 1 (Pa1), paddle speed ratio (Rat) and deaeration (Dea). Interactions are not shown for reasons of clarity.

optimization. Therefore, we deliberately screened for a powder with a challenging flow behaviour. As segregation and lubrication were not the focus of current study, a pure material that did not require lubrication was selected to reduce possible confounding. Based on preliminary trials with various fillers, a specific MCC grade (Vivapur® 101) was selected as it exhibited the required properties for this direct compression study. This material could be processed without lubrication during long-term preliminary trials and was characterized by a poor flowability and high compressibility. Van Snick et al. introduced a material database for excipients and blends in the context of direct compression (Van Snick et al., 2018). Herein, Vivapur® 101 was included to compare its material properties with those of other excipients and excipient-API blends. Interestingly, the powder shows similar flow and compressibility as blends of various APIs (10% w/w) with a poorly flowing MCC grade (Avicel pH 101) making Vivapur® 101 an excellent model powder for our experimental study since it bears some predictability for (low dosed) API blends with a poorly flowing MCC-filler.

3.2. Design analysis

Not all runs of the experimental design were successfully completed depending on the specific combination of paddle design and process-settings (Table 1). Run 12 and 35 which were performed using paddle type C at high paddle and low turret speeds resulted in extremely high tablet weight variabilities and incomplete die-filling. This phenomenon was explained by the specific paddle design applied at high paddle speed. Comparing the paddle designs illustrated in Fig. 2, the fine cylindrical paddles used in paddle type C have a limited paddle surface available for powder relocation. The latter design feature combined with a high paddle wheel speed seriously limited the transport of powder towards the dies since the paddles are rather slicing through the powder bed instead of transporting material to the dies. Moreover, at low turret speed there is less contribution of suction fill, which is expected to be an important mechanism for the die-filling of more aerated powder beds. Suction fill is a result of the partial vacuum created under the powder by the rapid withdrawal of the lower punch at the overfill station (Jackson et al., 2007). Due to the specific implementation of the cylindrical paddles at different heights in paddle wheel C, more agitation and thus aeration of the powder bed inside the forced feeder was expected compared to the other paddle designs where all paddles were implemented at the same height. This all resulted in a powder bed which experienced less forces to transport powder to the dies.

Runs 16, 19, 36, 38 and 39 were also excluded from the experimental design as feed frame blockage occurred quickly after the start of the experiments. These runs were typified by the use of paddle type D at low paddle speed. Analysis of the feed frame after obstruction revealed that the scrapers included in this design were the root cause of the problems for these runs (Fig. 2). Dense compacts of powder were noticed in the region close to the scrapers, which produced a high counterforce for the paddle wheels during rotation. The presence of scrapers in the powder stream was problematic for MCC, which is well known for its high compressibility and hence ability to easily form compacts. This effect mainly manifested at low paddle speed (i.e. low paddle force). No obstructions were noticed at higher paddle speeds indicating that a certain force from the paddles is necessary to overcome this dead zone inside the powder flow. The full data set is provided in Table 1 of the Supplementary Information together with an overview of the model quality for each response in Fig. 2 of the Supplementary Information.

3.3. Impact of process-settings on die-filling

An overview on how feed frame design and process settings influenced the mean TW, and thus the die-filling step, is given by the coefficient plot in Fig. 3. The regression coefficient of a specific factor represents the quantitative change in response value when this factor is increased from its average to its high level, keeping all other factors at their average value. As expected, the mean TW significantly decreased when tableting at higher speeds. This is a well-known phenomenon which can be explained by the reduced filling time of each die at higher tableting speed since it is exposed for a shorter time to the powder bed. The positive impact of higher overfill levels on the mean TW was expected as well, since overfilling is generally applied during conventional compaction processes to ensure a consistent die-filling. It relies on the fact that more powder is filled into the die-cavities than required while the excess of powder is expelled from the die into the feeder further down the turret (i.e. dosing cam) by raising the lower punch back to the desired fill depth (Kirsch, 2015).

When analysing the process-settings related to the forced feeder, an overall higher mean TW was observed with increasing speeds of the 'feeding' paddle wheel (Pa1). This reflected the contribution of forced feeding to die-filling in rotary tablet presses, a feeding mechanism that is expected to be crucial for poorly flowing formulations. At higher Pa1

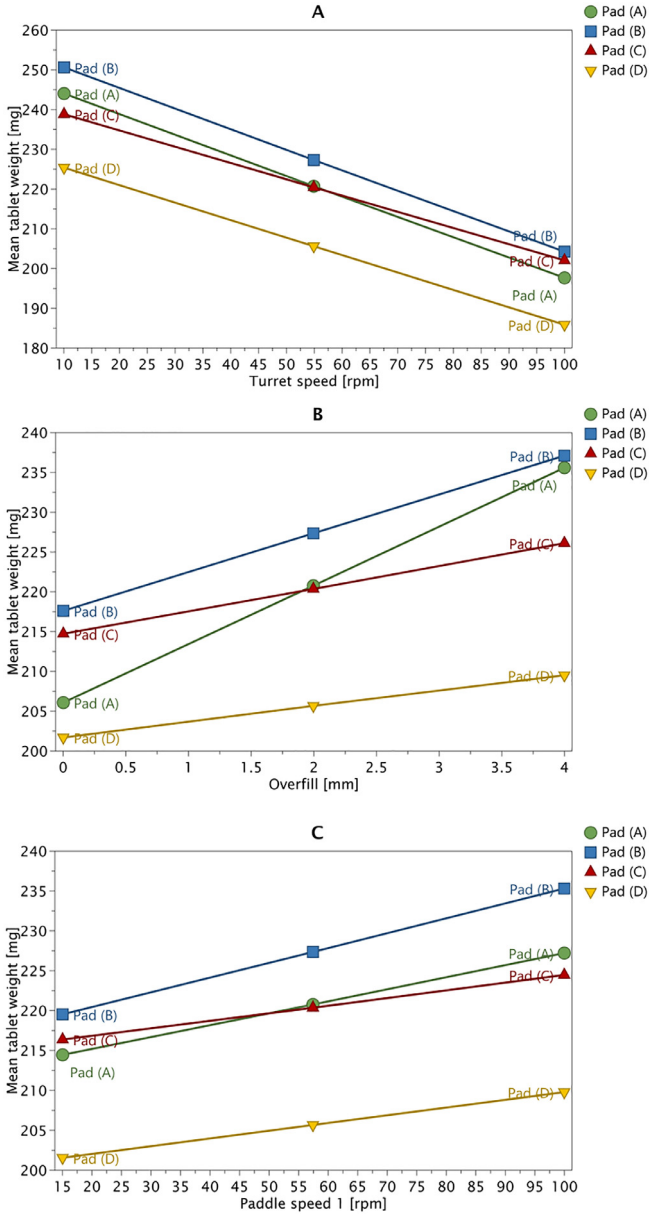


Fig. 4. Predicted changes in mean TW due to interaction of paddle design (see legend) with turret speed (A), overfill (B) and speed of Pa1 (C). The most relevant interactions are shown.

speed, the powder bed is exposed to more paddle passes during the short exposure time, pushing more powder to the dies. Despite the latter effect of Pa1, no significant impact was noticed when increasing the speed of the metering wheel (Pa2) relatively to that of Pa1. Similar findings were previously reported for a powder with excellent flow properties, suggesting that this effect is independent of the formulation's flow properties (Peeters et al., 2015). The settings of many rotary tablet presses by various manufacturers use by default a higher speed of Pa2 (i.e. Pa2/Pa1 ratio > 1), but some tablet presses do not allow the speed of both paddle wheels to be altered independently from one another, leaving no other choice for the operator than to perform experiments at the default settings. However, the impact of (needlessly) applying higher paddle speeds on the quality attributes of tablets (e.g. hardness and dissolution rate) might be significant due to the possible effects of overlubrication related to more shear on the powder bed in the feed frame (Mehrotra et al., 2007; Mendez et al., 2012; Peeters et al., 2016). Taking this information into account, there are sufficient arguments to question this default setting of paddle speeds (i.e. Pa2/Pa1 ratio > 1) and we would recommend to step away from this approach if possible.

The motivation to use the deaeration opening, which is a standard feature of the feed frame as represented in Fig. 2, was limited for this investigated formulation. Only at high speeds of Pa2, there was a positive but small effect on the mean TW. Moreover, some runs with paddle type C and an active deaeration opening experienced material loss from the feed frame through this opening. This observation at specific tableting conditions in combination with the limited impact at high paddle speed ratio does not support the use of this deaeration opening in future studies. However, it cannot be completely excluded that some specific formulations might benefit from this feature.

3.4. Impact of paddle design on die filling

The overall impact of the paddle designs (A–D) on the mean TW is presented in Fig. 3. Differences in powder transportation by varying paddle design can be directly correlated with this response as it reflects the amount of material that is filled in the dies. Therefore, the impact on mean TW is initially described to provide a primary understanding before focussing more in detail on the die-filing variability – a more complex response- to select an optimal paddle design. Experiments performed with paddle types A and B generally resulted in tablets with higher weight, while no significant impact of paddle type C was noticed in comparison to the impact of the other paddle types on this response. Interestingly, the use of paddle type D resulted in tablets with an overall lower mean TW. In order to better understand the contributions to these effects, interaction plots were analysed as illustrated in Fig. 4 for the most relevant interactions. An interaction plot predicts the change in a response (i.e. mean TW) due to interaction of the investigated factors (i.e. paddle design with the most relevant process settings).

For all paddle designs investigated in this study there was a negative effect of increasing turret speed on the die-filling step, since less time was available for die-filling (Fig. 4; A). Nonetheless, when using paddle type C, the effect of turret speed was less pronounced as evidenced by the lower slope. This is probably due to the specific design of type C, creating a more aerated powder bed which is inevitably more prone to a facilitated die-filling by suction fill and vibrations, two factors that are correlated with the turret speed (Jackson et al., 2007). A similar mean TW compared to paddle designs A and B can therefore be reached with paddle type C at high tableting speed despite its inferior die-filling performance at low turret speeds. However, for a comprehensive assessment of the benefits of each specific paddle design it is essential to include other process settings besides turret speed.

Overfilling of the dies had a positive impact on the mean TW for all investigated designs, but showed a dominant effect for the curved paddles (Fig. 4; B). Curved paddles without scrapers (i.e. type A) benefitted most from overfilling compared to the straight (i.e. type B) and the cylindrical paddle designs (i.e. type C) which is probably related to the radial curvature of the paddle, enabling a more efficient transport of the recovered powder by the metering wheel. When using scrapers in combination with the curved paddles (i.e. type D), the positive effect on die-filling by presenting more powder to the dies was largely counteracted by the obstruction and compaction of the powder at the scrapers in the feed frame.

Performing tableting experiments at a high Pa1 speed proved beneficial for most paddle designs except for paddle type C where a less significant change in mean TW was noticed (Fig. 4; C). This is also linked to the reduced paddle surface and a more aerated powder bed as described in a previous section. It is clear from these interaction plots that process settings and paddle design are associated with each other and need to be analysed together.

Besides the impact on mean TW, various other product and process responses are influenced by these factors. An overview is provided by performing PCA on the extensive data set, which resulted in the bi-plot

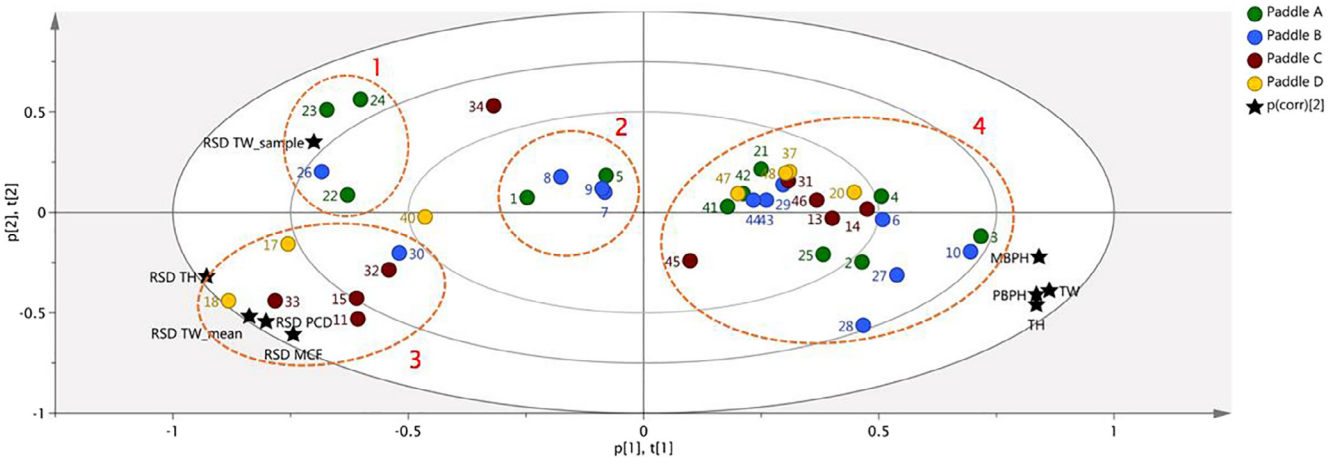


Fig. 5. PC_1 vs. PC_2 bi-plot combining the loadings (DOE-responses, black stars) with the scores (circles, colours in function of paddle design; see legend) for which the numbers represent the corresponding experimental run at the specific process settings. Clusters of scores are indicated and numbered by dashed ellipses.

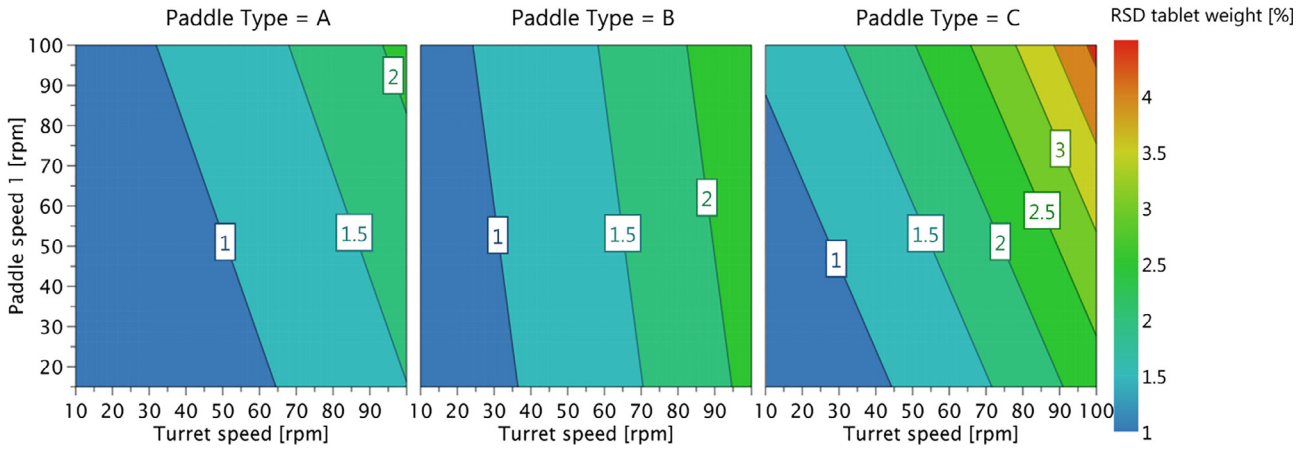


Fig. 6. Contour plots for die-filling variability (depicted by RSD_TW_mean) of three paddle designs (A, B and C) in interaction with paddle speed 1 (Pa_1) and turret speed at constant other CP process settings.

at Fig. 5. The model contains 2 principal components (PC) which accounted for 88.2% of variance in the dataset with PC_1 and PC_2 comprising 67.8% and 20.4%, respectively. The loadings on the bi-plot represent the investigated responses of the experimental design. A good correlation between tablet weight (TW), tablet hardness (TH) and the bottom punch heights at pre- and main compression (PBPH and MBPH) was observed at the right side of the origin in PC_1-direction. When switching the focus to the other side of the origin in PC_1-direction, a cluster was detected capturing the RSDs of the DOE-responses. This proved the validity of the concept on which control loops for adjustment of the tablet weight in rotary tablet presses are based (Peeters, 2014).

Moreover, the bi-plot can be used to interpret the obtained dataset of the experimental design. The RSDs of the responses represent the variability in the die-filling process and hence are representative for potential fluctuations in die-filling over tableting time. A robust tableting process is typified by low RSD-values since these guarantee a consistent die-filling. It is important to realize that in many cases a high die-filling variability during tableting is more problematic compared to a lower tablet weight because the correction mechanisms (e.g. turret speed) for lowering the variability are generally more invasive for the tableting process compared to the weight control by varying the fill depth. Therefore, it is important to have sufficient knowledge on how the design of the paddles and process settings can impact this variability before the start of the tableting process, to allow operators to select the optimal settings. Nevertheless, understanding of the physical impact of varying the paddle design - by analysis of the mean tablet weight- was important to reveal the underlying mechanisms in powder transport inside the feed frame and consequently die-filling variability.

The scores in the bi-plot (Fig. 5) represent the DOE-runs performed using specific paddle design and process settings. The main cause of variability is an increase in tableting speed, since all runs performed at 100 rpm are located left of the origin in PC_1-direction and the coefficient plot for tablet weight variability in the Supplementary Information (Fig. 3) confirms this impact. It was expected that performing experiments at high turret speed would increase the risk for variability on the die-filling. However, an important subdivision in degree of variability can be made for these high speed runs, For the experiments in cluster 1 (Fig. 5), there was mainly random variability in die-filling (i.e. higher RSD TW_{sample}) which did not result in fluctuations of mean TW over the tableting time (i.e. low RSD TW_{mean}).

When focusing on runs 22, 23, 24 and 26 (Table 1), these were performed with paddle types A and B at tableting conditions restricting efficient die-filling: a high tableting speed reduced the contact time for die-filling, no overfill limited the effect of suction fill and a low paddle speed limited forced feeding. Hence, die-filling mainly relied on gravimetric feeding which is known to be challenging at high tableting speeds, especially for poorly flowing powders (Jackson et al., 2007; Sinka et al., 2004; Wu et al., 2003). Interestingly, the random variability for these runs was easily bypassed using overfill as represented by cluster 2 in Fig. 5. The runs shifted towards the origin of the bi-plot, due to a strong reduction in the random die-filling variability and increase

of the mean TW. Therefore, the benefit of applying overfilling on die filling variability is clearly demonstrated, especially at high turret speeds.

The highest risk of variability in die-filling was observed for high speed runs with paddle types C and D (cluster 3, Fig. 5), especially when these paddles are operated at high paddle speeds. For paddle type C, the reason of this variability was already discussed in a previous section (i.e. paddles rather slicing through the more aerated powder bed). An important difference with paddle types A and B is that overfilling did not result in a significantly reduced die-filling variability for the runs performed with these paddles. The paddles of type C have a limited surface and were therefore unable to efficiently transport the excess of powder generated by the overfilling mechanism while for paddle type D more die-filling variability was probably induced due to the additional amount of powder reintroduced in the powder stream, hence increasing the risk for (random) compaction at the scrapers.

A large cluster (4) at the right side of the origin in PC1-direction was also distinguished, showing the centerpoint runs and the runs at lower turret speed which were considered as good quality runs (i.e. low TW_RSD values and high mean TW).

3.5. Selection of a paddle design

Based on the information about paddle design in combination with our understanding of the importance of limiting die-filling variability, this section was intended to select the most robust paddle design regarding weight variability in function of the process settings. The change in variability of die-filling (i.e. RSD TW_{mean}) for paddle types A, B and C during tableting at comparable ranges of process-settings are displayed by the contour plots in Fig. 6. Paddle type D was not taken into account since its design strongly resembled type A, while it resulted in less effective tableting runs regarding die-filling uniformity. When comparing the paddle designs in function of turret speed, the curved paddles (type A) stands out regarding their ability to withstand the unfavourable effects of turret speed on die-filling consistency. When using this paddle type, a $RSD_TW_{mean} < 1.5\%$ can be ensured over the whole range of turret speeds by additional fine-tuning of the paddle speed (i.e. lowering the paddle speed). This was not the case with the other two paddle designs, as these showed more weight variability at higher turret speeds. The beneficial effect of design A is most likely due to the specific curvature of the paddles, which are expected to transport and expose the powder to the dies in a more reproducible manner compared to other designs. Therefore, it can be stated that the most robust paddle design regarding die-filling consistency is a radial curved paddle (type A) which should therefore be preferentially used when working with a poorly flowing and cohesive formulation having similar material properties.

4. Conclusion

This research study revealed that the die-filling performance on a high speed rotary tablet press can be fine-tuned by a well-considered selection of feed frame design at its most optimal process settings. As the role of the rotary tablet press in a continuous tableting manufacturing line is steadily evolving towards a more agile unit-operation, capable of handling fluctuations in the upstream material flow, it becomes increasingly important to have more knowledge on how the feed frame design and process-settings such as tableting speed, paddle speed and overfill level might influence the tableting process. Turret speed is an obvious parameter of the rotary tablet press to adapt for variations in throughput, but nevertheless it is the factor that most significantly impacts the die-filling step of the tableting process. Moreover, it can be recommended to use overfilling, while there were no arguments to operate with deaeration or at paddle speed ratios higher than 1. In fact, the speed of the paddle wheel should be kept as low as possible regarding potential overlubrication and increased risk for high tablet weight variabilities. The approach used in this study enabled to compare various paddle designs for die-filling uniformity in function of varying tableting speeds, eventually resulting in the selection of one paddle type (i.e. type A) as the most robust paddle design for a poorly flowing formulation. This work proved that a proper selection of feed frame design can undeniably increase the flexibility of the rotary tablet press when this unit-operation must respond to variations in mass flow.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.ijpharm.2018.06.047.

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